

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1876

Red Hill 138 kV
July 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1876. The Interconnection Customer proposes a 72.8 MW solar photovoltaic facility interconnecting at the Red Hill 138 kV substation in the Virginia area. The request was received on September 03, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 477 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,573,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Grove Street – Charlottesville 138kV L
Contingency: Colgate Substation – Albreda Tans Mtn Tap 138kV
Loading: 125.7% (Rating: 240 MVA)
DFAX: 12.6%
Pre-project loading: 121.8%
2. Facility: Gordonsville – Proffit 138kV L
Contingency: Louisa Generation Facility – Fluvanna P.S 345kV
Loading: 108.9% (Rating: 364 MVA)
DFAX: 6.8%
Pre-project loading: 107.5%
3. Facility: Colgate – Proffit 138kV
Contingency: Louisa Generation Facility – Fluvanna P.S 345kV
Loading: 108.4% (Rating: 385 MVA)
DFAX: 6.8%
Pre-project loading: 107.2%
4. Facility: Colgate – Albreda Tans Mtn Tap 138kV
Contingency: Louisa Generation Facility – Fluvanna P.S 345kV
Loading: 107.6% (Rating: 429 MVA)
DFAX: 6.8%
Pre-project loading: 106.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grove Street – Charlottesville 138kV L	Rebuild line with higher-capacity conductor	\$21,964,000	6% (\$1,270,000)
Proffit Substation – Gordonsville 138kV	Reconductor line	\$3,199,000	4% (\$114,000)
Colgate – Proffit 138kV Line	Reconductor line	\$3,321,000	4% (\$118,000)
Colgate – Albreda Tans Mtn Tap 138kV	Reconductor line	\$1,992,000	4% (\$71,000)
	Total allocated to project		\$1,573,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.